Guilherme Duarte

Curriculum Vitae

Department of Computer Science
Faculdade de Ciências da Universidade do Porto
(+351) 910287588
guilherme.duarte@fc.up.pt
dblp page

Information

Full Name: Guilherme Manuel Carvalho de Melo Duarte.

Date of Birth: 25th of October 2000.

Nationality: Portuguese.

Research Interests

Formal Languages and Automata Theory, Combinatorial Analytics, Descriptional Complexity, Average-Case Complexity, Algorithms and Complexity Analysis.

Education

2024 – present **PhD in Computer Science**, *Faculdade de Ciências da Universidade do Porto, Portugal.*

2022 – 2024 **MSc in Computer Science**, *Faculdade de Ciências da Universidade do Porto, Portugal*, Specialisation in Reliable Computing.
Grade: 18/20

2019 – 2022 **BSc in Computer Science**, *Faculdade de Ciências da Universidade do Porto, Portugal*, Complementary Formation Centred in Computer Science.
Grade: 18/20

Publications

Conference and Workshop Papers

2025 **G. Duarte, N. Moreira, and R. Reis**, Two-Word Shuffle: Some Results, in *The 26th International Conference on Descriptional Complexity of Formal Systems, Loughborough, United Kingdom.*

2024 **G. Duarte, N. Moreira, L. Prigioniero, and R. Reis**, Operational Complexity on Block Languages, in *The 14th International Workshop on Non-Classical Models of Automata and Applications, Göttingen, Germany.*

G. Duarte, N. Moreira, L. Prigioniero, and R. Reis, Block Languages and their Bitmap Representations, in *The 28th International Conference on Implementation and Application of Automata, Akita, Japan.*

M. Cunha, G. Duarte, R. Andrade, R. Mendes, and J. P. Vilela, Privkit: A Toolkit of Privacy-Preserving Mechanisms for Heterogeneous Data Types, in *The 14th ACM Conference on Data and Application Security and Privacy, Porto, Portugal.*

G. Duarte, M. Cunha, and J. P. Vilela, A Privacy-Aware Remapping Mechanism for Location Data, in *The 39th ACM/SIGAPP Symposium On Applied Computing, Avila, Spain.*

Thesis

2024 **G. Duarte**, Minimal Automata and Operational State Complexity of Block Languages, MSc thesis.
Advised by Rogério Reis and Nelma Moreira
Grade: 20/20

2022 **G. Duarte**, Enhancing a Framework for Location Privacy, BSc thesis.
Advised by João P. Vilela and Mariana Cunha
Grade: 18/20

Grants

January, 2025 – present **PhD Studentship**, *Fundação para a Ciência e a Tecnologia*, 2024.01528.BD.
Departamento de Ciências de Computadores da Faculdade de Ciências da Universidade do Porto.

June, 2024 – December, 2024 **Research Grant Bachelor's Degree**, *Centro de Matemática da Universidade do Porto*, UIDB-00144-2020.
Departamento de Ciências de Computadores da Faculdade de Ciências da Universidade do Porto.

January, 2023 – June, 2023 **Research Fellowship**, *Centro de Informática e Sistemas da Universidade de Coimbra*, IT137-22-169.
Departamento de Engenharia Informática da Faculdade de Ciências e Tecnologia da Universidade de Coimbra.

March, 2022 – November, 2022 **Research Initiation Fellowship**, *Centro de Matemática da Universidade do Porto*, UIDB-00144-2020.
Departamento de Matemática da Faculdade de Ciências da Universidade do Porto.

Software

September, 2024 – present **Descriptional Complexity DataBase (DesCo)**, *khilas.dcc.fc.up.pt/desco*.
DesCo is a Web based information system for descriptional complexity results, currently focused on the operational complexity of regular and subregular languages.

September, 2023 – present **Tools for Formal Languages Manipulation (FAdo)**, *fado.dcc.fc.up.pt*.
The FAdo system provides an open source extensible high-performance software library for the symbolic manipulation of automata and other models of computation.

February, 2022 – September, 2023 **A Privacy Toolkit (Privkit)**, *privkit.fc.up.pt*, .
Privkit is a privacy toolkit for testing privacy-preserving mechanisms and to quantify the achieved privacy/utility level.

Positions of Responsibility

Local Arrangement Committee

June, 2024 **The 14th ACM Conference on Data and Application Security and Privacy**, Porto, Portugal.

May, 2022 **Final das Olimpíadas Nacionais de Informática**, Porto, Portugal.

Peer Review

2025 – present Conference on Descriptional Complexity of Formal Systems.
Workshop on Non-Classical Models of Automata and Applications.
Conference on Developments in Language Theory.

2024 – present Conference on Implementation and Application of Automata.

Activities

Presentation of Papers

2025 **Two-Word Shuffle: Some Results**, in *The 26th International Conference on Descriptional Complexity of Formal Systems, Loughborough, United Kingdom*.

2024 **Operational Complexity on Block Languages**, in *The 14th International Workshop on Non-Classical Models of Automata and Applications, Göttingen, Germany*.

A Privacy-Aware Remapping Mechanism for Location Data, in *The 39th ACM/SIGAPP Symposium On Applied Computing, Avila, Spain*.

Teaching

Fall 2025 **Programming I**, *CC1015*, in Faculdade de Ciências da Universidade do Porto.

Spring 2025 **Computational Models**, *CC1004*, in Faculdade de Ciências da Universidade do Porto.

Students

2024 – 2025 **Ana Sofia Teixeira**, *DesCo – A Descriptional Complexity Platform*, in Faculdade de Ciências da Universidade do Porto.
MSc co-advisor